Essential Summary

A Generalized Padeé Approximation Method for Solving Homoclinic and Heteroclinic Orbits of Strongly Nonlinear Autonomous Oscillators

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1 Background and Motivation

The classical Padeé approximation has been widely applied in nonlinear dynamics, with various researchers extending it by replacing polynomial components with exponential functions to better capture asymptotic homoclinic decay. However, these modifications were proposed case-by-case, without a **unifying theoretical framework**.

This paper (the English version of the GPA method) proposes an intrinsic extension: numerator and denominator are generalized from polynomials to **series of any function type**. This unifies all prior modifications and extends the method to **both homoclinic and heteroclinic orbits** across three fundamentally different oscillator classes: cubic–quintic Duffing, Φ^6 potential, and Duffing-harmonic.

2 Core Innovation

2.1 The Generalized Definition

Let $\widehat{H}_m = \{P : P(z) = \sum_{i=0}^{m} a_i g_i(z)^i\}$ extend the polynomial space H_m. The **Generalized Padeé Approximant** $\mathrm{GPA}[L/M]_f$ is $P_L/Q_M \in G(L,M)$ satisfying

$$f(z) - \frac{P_L(z)}{Q_M(z)} = \mathcal{O}(z^{L+M+1}). \tag{1}$$

All quasi-Padeé variants (Mikhlin 1995, Manucharyan & Mikhlin 2005, Zhang et al. 2011) become special cases.

2.2 Two Novel Approximants

For $\ddot{x} + g(x) = 0$ with saddle $H(h,0)$, the procedure: (1) derive power series coefficients a_i via substitution; (2) determine initial amplitude a from the energy integral $\int_h^a g(x)dx = 0$; (3) construct the GPA using hyperbolic functions that match the boundary conditions:

Homoclinic orbits ($x(\pm\infty) = h$):

$$\boxed{\mathrm{GPA}_{\mathrm{hom}}[L/M] = \frac{\sum_{i=0}^{L} \alpha_i \operatorname{sech}^i t}{1 + \sum_{i=1}^{M} \beta_i \operatorname{sech}^i t}}. \tag{2}$$

Heteroclinic orbits ($x(\pm\infty) = h_{1,2}$):

$$\boxed{\mathrm{GPA}_{\mathrm{het}}[L/M] = \frac{\displaystyle\sum_{i=0}^{L} \alpha_i \tanh^i t}{1 + \displaystyle\sum_{i=1}^{M} \beta_i \tanh^i t}}. \tag{3}$$

The $L + M + 1$ unknown coefficients $\{\alpha_i, \beta_i\}$ are found by Taylor-expanding the GPA around $t = 0$ and matching with the power series $\{a_i\}$. The hyperbolic basis functions automatically satisfy the asymptotic boundary conditions ($\mathrm{sech}^i\, t \to 0$ at $\pm\infty$; $\tanh^i t \to \pm 1$ at $\pm\infty$).

The coefficient-matching problem yields a system of $L+M$ linear equations (from Taylor matching) plus one nonlinear condition (from the energy integral), solved by standard numerical methods.

3 Key Results: Three Oscillator Classes

3.1 Cubic–Quintic Duffing Equation

$$\ddot{x} + c_1 x + c_3 x^3 + c_5 x^5 = 0, \qquad c_1 = 2,\ c_3 = -10,\ c_5 = 9. \tag{4}$$

Five fixed points (three centers, two saddles). Two homoclinic orbits around outer centers and one heteroclinic pair linking the two saddles coexist. GPA[4/4] for homoclinic and GPA[7/7] for heteroclinic capture all orbits accurately.

$$\int_{a_0} g(x)\,\mathrm{d}x = 0, \tag{17}$$

where h_i is the abscissa of saddle point. For heteroclinic orbit, the initial values $a_0 = 0$ and a_1 which can be determined by

$$\frac{1}{2}a_1^2 + k(0) = k(h_j), \tag{18}$$

where

$$k(x) = \int g(x)\,\mathrm{d}x,$$

h_j is the abscissa of any saddle point which is linked by the heteroclinic orbit. When $g(x)$ is a polynomial function and rational function, the parameters $a_i\,(i = 2, 3, \ldots)$ can always be obtained by substituting Eq. (15) into Eq. (14). The homoclinic solution should satisfy the following condition:

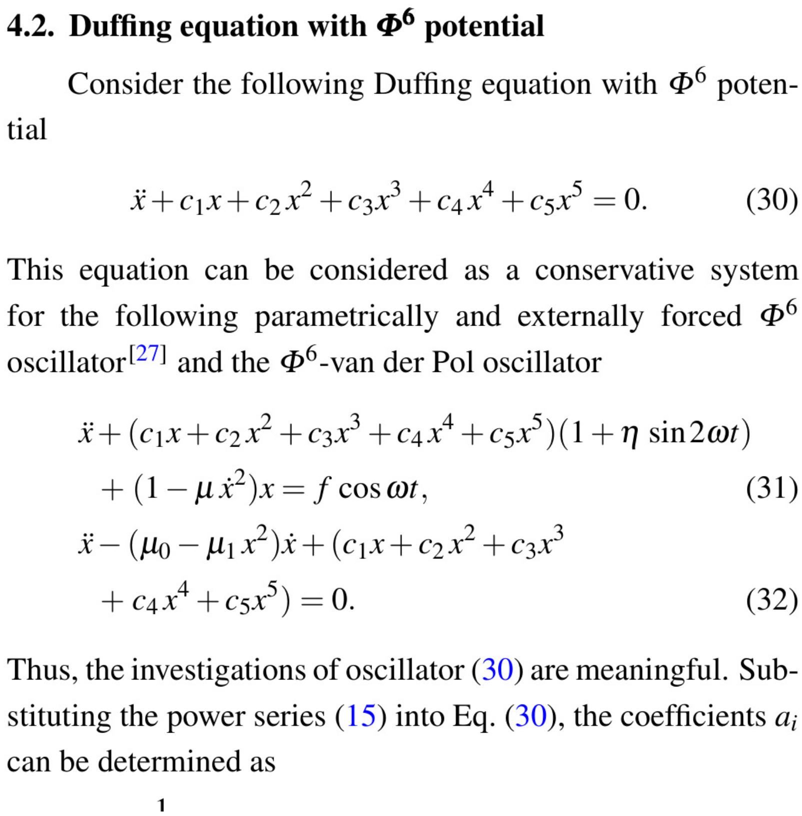
$$x(\pm\infty) = h_i. \tag{19}$$

Figure 1: **Figure 1.** (Original Fig. 1) Homoclinic and heteroclinic orbits of the cubic–quintic Duffing oscillator. Solid: Runge–Kutta. Dotted: GPA method. The quasi-Padeé solution of Feng et al. is also shown for comparison.

3.2 Duffing Equation with Φ^6 Potential

$$\ddot{x} + c_1 x + c_2 x^2 + c_3 x^3 + c_4 x^4 + c_5 x^5 = 0. \tag{5}$$

With $c_1 = c_3 = c_5 = 1, c_2 = c_4 = -1.5$, three saddle points enable both homoclinic and heteroclinic orbits simultaneously. GPA[5/5] and GPA[7/7] are compared with classical Padeé approximants and the Runge–Kutta method.



4.2. Duffing equation with Φ^6 potential

Consider the following Duffing equation with Φ^6 potential

$$\ddot{x} + c_1 x + c_2 x^2 + c_3 x^3 + c_4 x^4 + c_5 x^5 = 0. \tag{30}$$

This equation can be considered as a conservative system for the following parametrically and externally forced Φ^6 oscillator[27] and the Φ^6-van der Pol oscillator

$$\begin{aligned}&\ddot{x} + (c_1 x + c_2 x^2 + c_3 x^3 + c_4 x^4 + c_5 x^5)(1 + \eta \sin 2\omega t)\\&\quad + (1 - \mu \dot{x}^2)x = f \cos \omega t, \end{aligned}\tag{31}$$

$$\begin{aligned}&\ddot{x} - (\mu_0 - \mu_1 x^2)\dot{x} + (c_1 x + c_2 x^2 + c_3 x^3\\&\quad + c_4 x^4 + c_5 x^5) = 0.\end{aligned}\tag{32}$$

Thus, the investigations of oscillator (30) are meaningful. Substituting the power series (15) into Eq. (30), the coefficients a_i can be determined as

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Figure 2: **Figure 2.** (Original Fig. 3) Homoclinic and heteroclinic orbits of the Φ^6 oscillator. Solid: Runge–Kutta. Dotted: GPA method. Dashed: classical Padeé approximant. The GPA achieves demonstrably higher accuracy at the same approximation order.

3.3 Generalized Duffing–Harmonic Oscillator

$$\ddot{x} + \frac{\lambda x + \mu x^3}{1 + \nu x^2} = 0. \tag{6}$$

This rational-restoring-force oscillator is a standard benchmark. With $\lambda = -1, \mu = 5, \nu = 5$, GPA[3/3] captures two homoclinic orbits. With $\lambda = 3, \mu = -5, \nu = 2$, GPA[8/8] captures heteroclinic orbits. The rational restoring force introduces no additional difficulty.

4 Significance and Impact

1. **Unifying framework:** The GPA definition encompasses the classical Padeé approximant and all quasi-Padeé variants as special cases, providing a common theoretical language.

2. **First simultaneous homoclinic–heteroclinic treatment:** The $\operatorname{sech}^i t$ / $\tanh^i t$ dual approximant construction provides a unified pipeline for both orbit types — a capability absent from prior Padeé-based methods.

by Feng *et al.*[27] via classical Padé approximant are shown in Fig. 3. Meanwhile, the solutions of this equation via approximant (7) with the same approximation orders are also shown in Fig. 3. From this figure, we can find that the solutions obtained by the present method are very close to the numerical ones, and the accuracies of our solutions are higher than those of the solutions obtained via approximant (7) and the solutions given by Feng *et al.*

$$a_2 = \frac{-\lambda a_0 - \mu a_0^3}{2\left(1+\nu a_0^2\right)},$$

$$a_3 = \frac{-\lambda a_1 - 3\mu a_0^2 a_1 + \lambda \nu a_0^2 a_1 - \mu \nu a_0^4 a_1}{6\left(1+\nu a_0^2\right)^2},$$

$$a_4 = \frac{a_0\left\{\lambda^2 + 3\mu^2 a_0^4 + \mu^2 \nu a_0^6 - 6(\mu - \lambda\nu)\, a_1^2 -\right.}{}$$

Figure 3: **Figure 3.** (Original Fig. 4) Homoclinic orbits of the Duffing-harmonic oscillator ($\lambda = -1, \mu = 5, \nu = 5$). Solid: Runge–Kutta. Dotted: GPA[3/3].

$$+\, a_0^4(27\mu^2 - 28\lambda\mu\nu + 7\lambda^2\nu^2 + 6\nu^2(-\mu +$$

$$\cdots.$$

When $\lambda = -1$, $\mu = 5$, and $\nu = 5$, equation (38) has two homoc procedure of generalized Padé approximation method, the homocl

$$GPA[3/3] = \pm\frac{1.26143\text{sech}(t) + 6.18}{1 + 4.91221\text{sech}(t) + 14}$$

When $\lambda = 3$, $\mu = -5$, and $\nu = 2$, equation (38) has a pair of heteroclinic orbits. Letting $N = M = 8$ in Eq. (22) and through the procedure of generalized Padé approximation method, the heteroclinic orbits of Eq. (38) can be obtained as

Figure 4: **Figure 4.** (Original Fig. 5) Heteroclinic orbits of the Duffing-harmonic oscillator ($\lambda = 3, \mu = -5, \nu = 2$). Solid: Runge–Kutta. Dotted: GPA[8/8].

3. **Superior accuracy vs. classical Padeé:** At the same approximation order, the hyperbolic-function GPA achieves demonstrably higher accuracy than polynomial-based Padeé approximants, especially for strongly nonlinear cases where polynomial bases struggle with exponential decay.

5 Limitations

- Conservative ($\varepsilon = 0$) systems only; damped systems require separate methods.
- Approximation order chosen empirically; no systematic error bound.
- Coefficient matching involves $L + M + 1$ nonlinear equations that may become ill-conditioned at high orders.

6 Connection to the GHFP/MGHFP Series

The GPA method and the GHFP/MGHFP family represent two **complementary methodological families** for strongly nonlinear oscillators. Both share the mathematical toolkit of rational function approximation, but pursue different routes: GPA constructs homoclinic/heteroclinic orbits directly via hyperbolic-function rational approximants in the time domain, while GHFP/MGHFP uses perturbation theory in the angular domain to derive amplitude–parameter relationships and bifurcation thresholds. The $\operatorname{sech}^i t$ basis introduced here also inspired the $x = a\sin^2\varphi + b$ homoclinic construction in the GHFP framework (JSV 2013).

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